

OASIS INFOBYTE

DATA ANALYTICS INTERNSHIP

LEVEL 1 – TASK 2

CUSTOMER SEGMENTATION ANALYSIS

Final Project Report

Submitted By

Madala Sai Sprisha

Internship

OASIS INFOBYTE – Data Analytics Internship

Domain

Data Analytics

Project Title

Level 1 – Task 2: Customer Segmentation Analysis

Technologies Used

Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn • Jupyter Notebook

Modules Implemented

Data Cleaning • Customer Aggregation • RFM Analysis • Feature Standardization • K-Means
Clustering • Customer Profiling • Segment Recommendations

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Year

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the OASIS INFOBYTE internship program for providing me with the opportunity to work on Level 1 – Task 2, Customer Segmentation Analysis. This project provided practical exposure to data cleaning, customer-level aggregation, RFM analysis, feature standardization, clustering, visualization, and business interpretation using Python.

I would also like to thank the mentors and coordinators for providing the project requirements and learning resources. The structured task helped me understand how customer transaction data can be transformed into meaningful behavioral segments and how those segments can support targeted marketing decisions.

During this project, I worked with the Online Retail dataset containing 541,909 transaction records and 8 variables. The implementation included duplicate removal, cancellation filtering, validation of Quantity and UnitPrice, handling of missing CustomerID values, creation of TotalPrice, construction of customer-level purchase metrics, RFM analysis, K-Means clustering, cluster profiling, visualization, and marketing recommendations.

Finally, I thank everyone who directly or indirectly supported the successful completion of this project. The experience strengthened my practical understanding of Python-based customer analytics and demonstrated how unsupervised learning can support data-driven customer relationship strategies.

ABSTRACT

This project focuses on Customer Segmentation Analysis of online retail transaction data using Python. The main objective is to clean transaction records, summarize customer purchasing behavior, calculate Recency, Frequency, and Monetary (RFM) measures, and apply K-Means clustering to identify distinct customer groups.

The input Online Retail dataset contains 541,909 records and 8 variables. The data preparation stage removed 5,268 duplicate records, 9,251 cancelled transactions, invalid Quantity and UnitPrice records, and transactions without CustomerID. After cleaning, 392,692 valid customer-linked transactions remained. Customer-level analysis produced 4,338 unique customers.

RFM features were calculated from transaction history. Recency represents the number of days since the customer's most recent purchase, Frequency represents the number of unique invoices, and Monetary represents total purchase value. These features were standardized using StandardScaler and evaluated using the Elbow Method. Four clusters were selected and implemented using K-Means.

The resulting segments were Regular Customers, At-Risk / Inactive Customers, Loyal High-Value Customers, and VIP / High-Value Customers. The largest group was Regular Customers with 3,054 customers, while the VIP / High-Value group contained only 13 customers but had the highest average frequency and monetary value. Segment-specific marketing recommendations were developed to support re-engagement, loyalty, retention, and personalized customer treatment.

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

The Customer Segmentation Analysis project focuses on transforming online retail transaction data into customer-level behavioral information and meaningful customer segments. The implementation begins with dataset loading and inspection, followed by data cleaning and validation. Customer transactions are then aggregated to calculate purchase behavior and RFM metrics.

The project uses Pandas and NumPy for data manipulation and numerical operations, Matplotlib and Seaborn for visualization, and Scikit-learn for feature standardization and K-Means clustering. The analysis is performed in a Jupyter Notebook environment.

1.2 Problem Statement

Raw retail transaction data contains individual invoices and product-level records, which makes it difficult to directly understand customer value and engagement. The problem addressed in this project is to transform transaction-level records into customer-level behavioral measures and group customers with similar purchasing patterns.

The resulting segments can help a business distinguish regular customers from at-risk customers and identify high-value customers who require stronger retention and loyalty strategies.

1.3 Project Objectives

1. Load and inspect the Online Retail transaction dataset.
2. Identify missing values and duplicate records.
3. Remove cancelled transactions and invalid Quantity or UnitPrice values.
4. Handle transactions without CustomerID for customer-level analysis.
5. Create TotalPrice and customer-level purchase metrics.
6. Calculate Recency, Frequency, and Monetary (RFM) features.
7. Standardize the RFM features before clustering.
8. Use the Elbow Method to determine a suitable number of clusters.
9. Apply K-Means clustering and profile the resulting customer groups.
10. Create customer segment summaries, visualizations, and marketing recommendations.

CHAPTER 2: DATASET & TOOLS

2.1 Dataset Description

The project uses the Online Retail.xlsx dataset. The original dataset contains 541,909 transaction records and 8 variables: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. These fields support transaction, product, date, customer, and geographic analysis.

2.2 Tools & Technologies Used

Tool / Technology	Purpose
Python	Programming and customer analytics
Pandas	Data loading, cleaning, grouping and aggregation
NumPy	Numerical operations
Matplotlib	Customer segmentation visualizations
Seaborn	Statistical/data visualization support
Scikit-learn	StandardScaler and K-Means clustering
Jupyter Notebook	Interactive implementation and documentation
Excel Dataset	Input Online Retail transaction data

2.3 Project Folder Structure

The project was organized into separate folders for the dataset, notebook implementation, output file, report documentation, and screenshots. The structure keeps the implementation, evidence images, and generated customer segment summary organized.

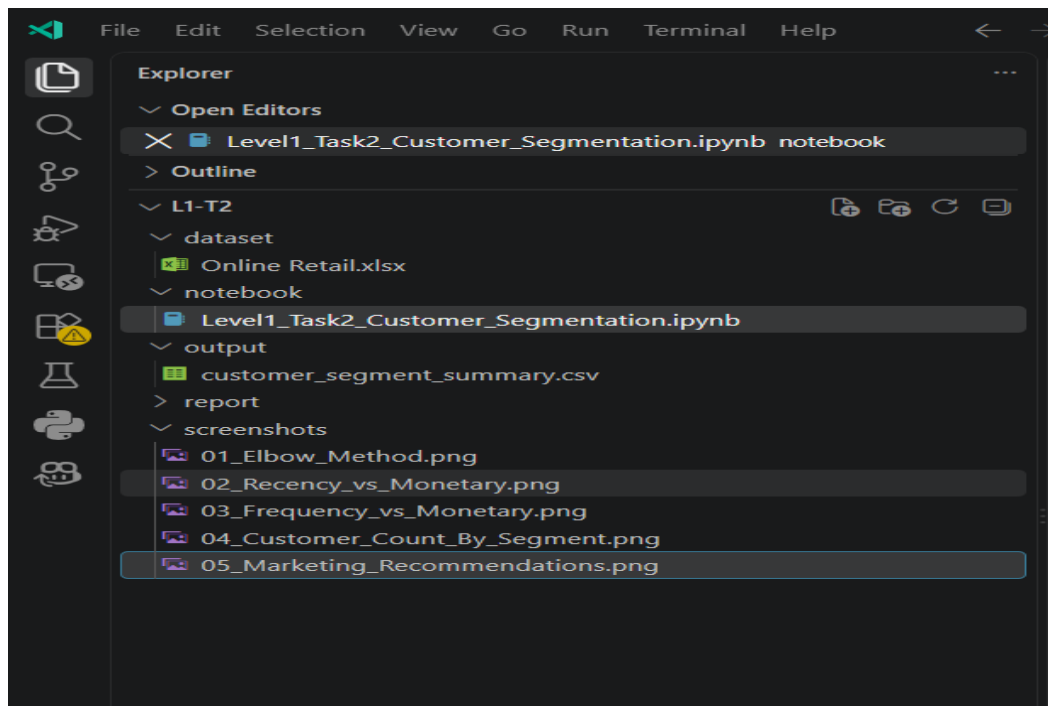


Figure 2.3: Project Folder Structure of the Customer Segmentation Analysis

2.4 Input Dataset

The notebook loads the Excel input file using the relative path `../dataset/Online Retail.xlsx`. The dataset covers transactions from 1 December 2010 to 9 December 2011. The InvoiceDate field is used to determine customer recency, while InvoiceNo and TotalPrice support frequency and monetary calculations.

	A	B	C	D	E	F	G	H	I	J	K	L
	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country				
1	536365	85123A	WHITE HANGING HE	6	01-12-2010 08:26	2.55	17850	United Kingdom				
2	536365	71053	WHITE METAL LANT	6	01-12-2010 08:26	3.39	17850	United Kingdom				
3	536365	84406B	CREAM CUPID HEAR	8	01-12-2010 08:26	2.75	17850	United Kingdom				
4	536365	84029G	KNITTED UNION FLA	6	01-12-2010 08:26	3.39	17850	United Kingdom				
5	536365	84029E	RED WOOLLY HOTTI	6	01-12-2010 08:26	3.39	17850	United Kingdom				
6	536365	22752	SET 7 BABUSHKA NE	2	01-12-2010 08:26	7.65	17850	United Kingdom				
7	536365	21730	GLASS STAR FROSTE	6	01-12-2010 08:26	4.25	17850	United Kingdom				
8	536366	22633	HAND WARMER UNI	6	01-12-2010 08:28	1.85	17850	United Kingdom				
9	536366	22632	HAND WARMER REC	6	01-12-2010 08:28	1.85	17850	United Kingdom				
10	536367	84879	ASSORTED COLOUR	32	01-12-2010 08:34	1.69	13047	United Kingdom				
11	536367	22745	POPPY'S PLAYHOUSE	6	01-12-2010 08:34	2.1	13047	United Kingdom				
12	536367	22748	POPPY'S PLAYHOUSE	6	01-12-2010 08:34	2.1	13047	United Kingdom				
13	536367	22749	FELTCRAFT PRINCES	8	01-12-2010 08:34	3.75	13047	United Kingdom				
14	536367	22310	IVORY KNITTED MUK	6	01-12-2010 08:34	1.65	13047	United Kingdom				
15	536367	84969	BOX OF 6 ASSORTED	6	01-12-2010 08:34	4.25	13047	United Kingdom				
16	536367	22623	BOX OF VINTAGE JIG	3	01-12-2010 08:34	4.95	13047	United Kingdom				
17	536367	22622	BOX OF VINTAGE AL	2	01-12-2010 08:34	9.95	13047	United Kingdom				
18	536367	21754	HOME BUILDING BL	3	01-12-2010 08:34	5.95	13047	United Kingdom				
19	536367	21755	LOVE BUILDING BLC	3	01-12-2010 08:34	5.95	13047	United Kingdom				
20	536367	21777	RECIPE BOX WITH M	4	01-12-2010 08:34	7.95	13047	United Kingdom				
21	536367	48187	DOORMAT NEW ENI	4	01-12-2010 08:34	7.95	13047	United Kingdom				
22	536368	22960	JAM MAKING SET W	6	01-12-2010 08:34	4.25	13047	United Kingdom				
23	536368	22913	RED COAT RACK PAR	3	01-12-2010 08:34	4.95	13047	United Kingdom				
24	536368	22912	YELLOW COAT RACK	3	01-12-2010 08:34	4.95	13047	United Kingdom				
25	536368	22914	BLUE COAT RACK PA	3	01-12-2010 08:34	4.95	13047	United Kingdom				
26	536369	21756	BATH BUILDING BLC	3	01-12-2010 08:35	5.95	13047	United Kingdom				
27	536370	22728	ALARM CLOCK BAKE	24	01-12-2010 08:45	3.75	12583	France				
28	536370	22727	ALARM CLOCK BAKE	24	01-12-2010 08:45	3.75	12583	France				

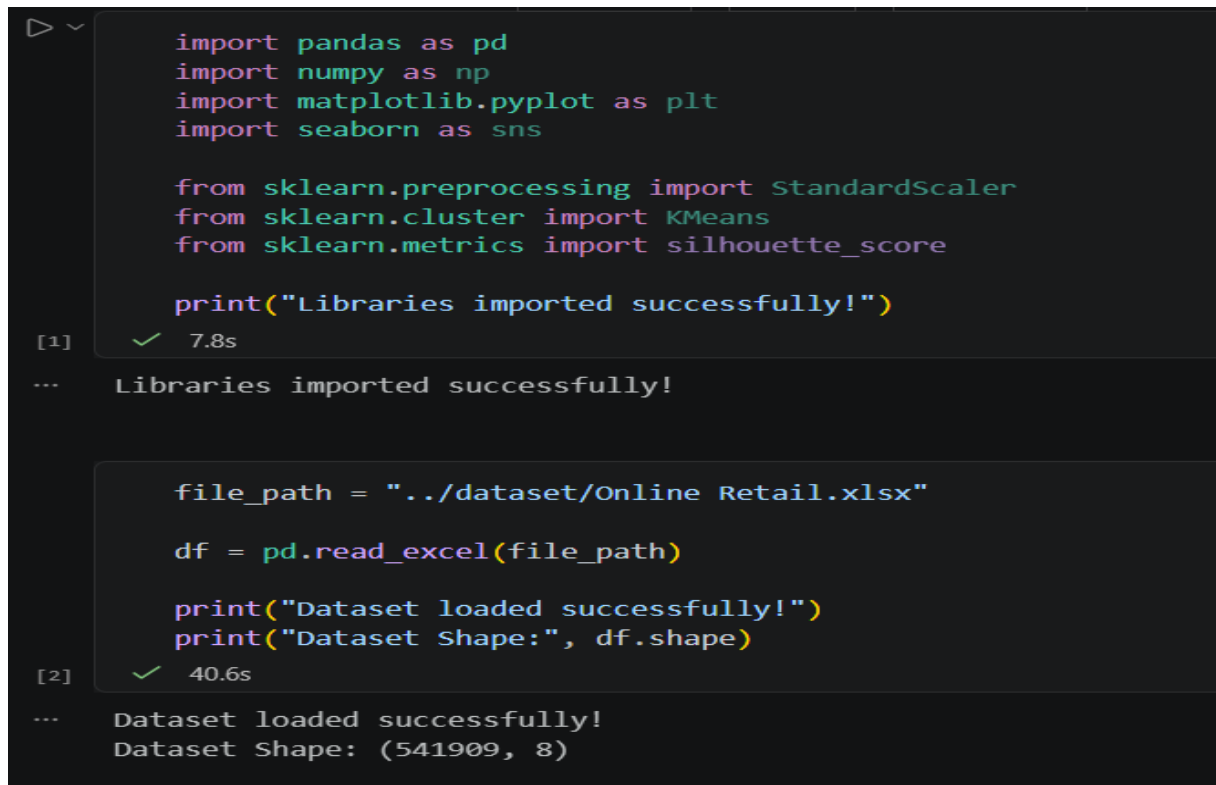
Figure 2.4: Input Online Retail Dataset

CHAPTER 3: CUSTOMER SEGMENTATION

IMPLEMENTATION

3.1 Library Import & Dataset Loading

The required Python libraries were imported successfully. Pandas was used for DataFrame operations, NumPy for numerical support, Matplotlib and Seaborn for visualization, and Scikit-learn for StandardScaler and K-Means. The Excel dataset was loaded into a Pandas DataFrame.



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

print("Libraries imported successfully!")

[1] ✓ 7.8s
... Libraries imported successfully!

file_path = "../dataset/Online Retail.xlsx"

df = pd.read_excel(file_path)

print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)

[2] ✓ 40.6s
... Dataset loaded successfully!
Dataset Shape: (541909, 8)
```

Figure 3.1: Library Import and Dataset Loading Implementation

Interpretation:

The required Python libraries were imported successfully, and the Online Retail dataset was loaded into a Pandas DataFrame without errors. The dataset initially contained 541,909 transaction records and 8 columns, providing the necessary transaction-level information for customer segmentation analysis.

3.2 Data Inspection & Missing Values

The dataset was inspected using `head()`, `shape`, column names, data types, and missing-value counts. The inspection showed that Description contained 1,454 missing values and CustomerID contained 135,080 missing values. CustomerID was especially important because customer-level segmentation requires a valid customer identifier.

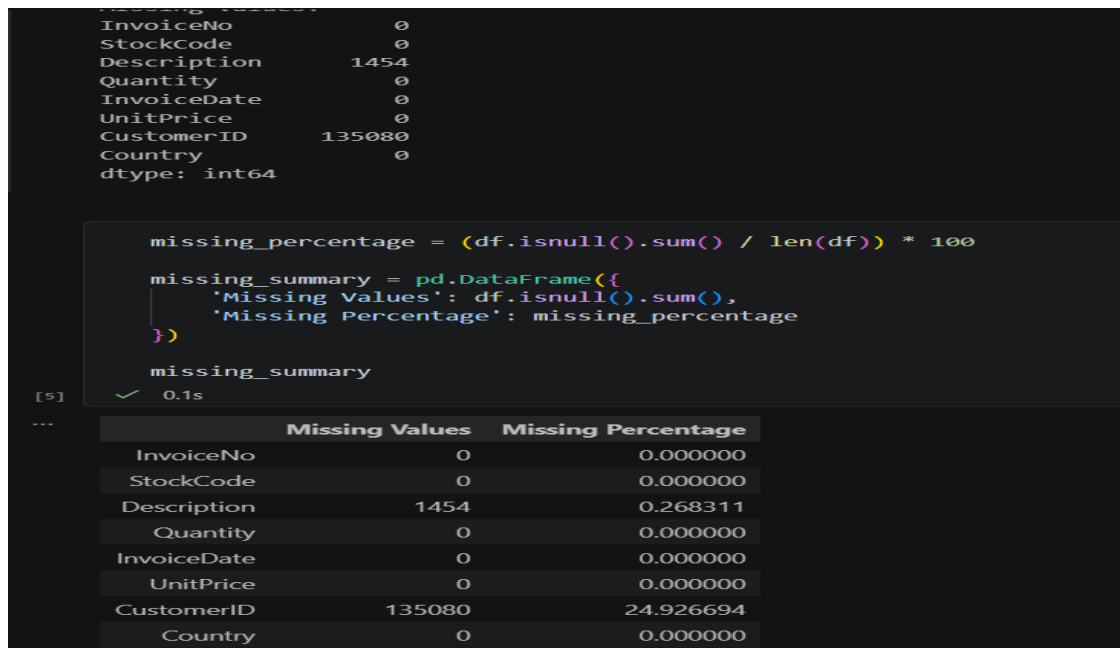


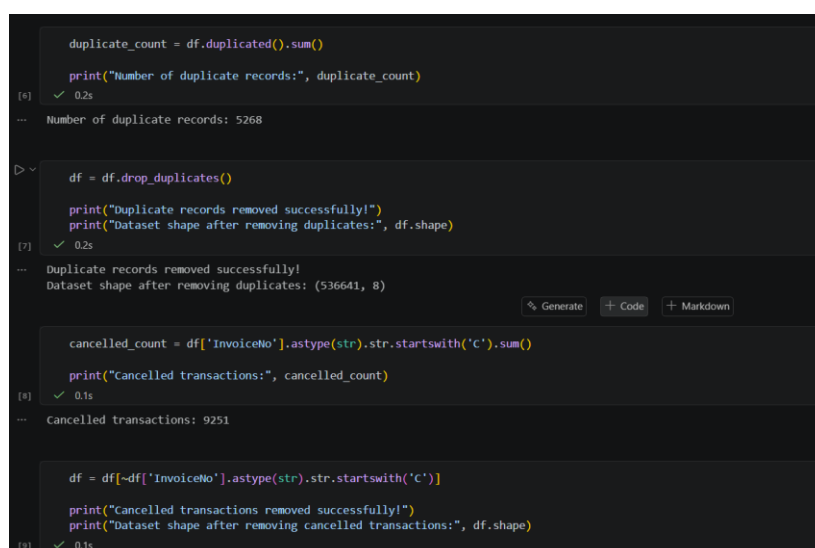
Figure 3.2: Dataset Inspection and Missing Value Analysis

Interpretation:

The inspection identified missing values in the dataset. The Description column contained 1,454 missing values, while CustomerID contained 135,080 missing values. Since CustomerID is essential for customer-level analysis, records without a valid customer identifier required appropriate handling before segmentation.

3.3 Data Cleaning

Data cleaning was performed in several stages. First, duplicate records were identified and removed. The dataset contained 5,268 duplicate records. Next, cancelled transactions were identified using InvoiceNo values beginning with 'C'; 9,251 cancelled transactions were removed.



```
[12] ✓ 0.0s
... Invalid Quantity and UnitPrice records removed successfully!
Dataset shape after cleaning: (524878, 8)

print("Minimum Quantity:", df['Quantity'].min())
print("Minimum UnitPrice:", df['UnitPrice'].min())
[13] ✓ 0.0s
... Minimum Quantity: 1
Minimum UnitPrice: 0.001

print("Missing CustomerID before cleaning:",
      df['CustomerID'].isnull().sum())
[14] ✓ 0.0s
... Missing CustomerID before cleaning: 132186

df = df.dropna(subset=['CustomerID'])
print("Missing CustomerID records removed successfully!")
print("Missing CustomerID after cleaning:",
      df['CustomerID'].isnull().sum())
print("Dataset shape after CustomerID cleaning:", df.shape)
[15] ✓ 0.0s
... Missing CustomerID records removed successfully!
Missing CustomerID after cleaning: 0
Dataset shape after CustomerID cleaning: (392692, 8)
```

Figure 3.3: Duplicate Record Detection and Removal

Interpretation:

The data-cleaning process identified **5,268 duplicate records**, which were removed to prevent repeated transactions from affecting customer-level calculations. Cancelled transactions were also identified using InvoiceNo values beginning with “C” and removed from the analysis. These steps improved the reliability of the transaction data used for subsequent customer segmentation.

3.4 Customer-Level Aggregation

After cleaning, transactions were grouped by CustomerID to create customer-level purchase measures. Purchase Frequency was calculated as the number of unique invoices, Purchase Value as the sum of TotalPrice, and Total Quantity as the total number of items purchased.

```
customer_summary.head()
[17] ✓ 0.0s
...
CustomerID Purchase_Frequency Purchase_Value Total_Quantity
12346.0 1 77183.60 74215
12347.0 7 4310.00 2458
12348.0 4 1797.24 2341
12349.0 1 1757.55 631
12350.0 1 334.40 197

customer_summary['Average_Purchase_Value'] = (
    customer_summary['Purchase_Value'] /
    customer_summary['Purchase_Frequency']
)
customer_summary.head()
[18] ✓ 0.0s
...
CustomerID Purchase_Frequency Purchase_Value Total_Quantity Average_Purchase_Value
12346.0 1 77183.60 74215 77183.600000
12347.0 7 4310.00 2458 615.714286
12348.0 4 1797.24 2341 449.310000
12349.0 1 1757.55 631 1757.550000
12350.0 1 334.40 197 334.400000
```

Figure 3.4: Customer-Level Purchase Aggregation

Interpretation:

The cleaned transaction data was grouped by CustomerID to generate customer-level purchase measures. Purchase frequency was calculated using unique invoices, purchase value was calculated from TotalPrice, and total quantity represented the number of items purchased. This aggregation transformed transaction-level records into customer-level behavioral information suitable for RFM analysis.

3.5 RFM Analysis

RFM analysis was used to represent customer behavior through three dimensions. Recency measures how recently a customer purchased, Frequency measures how often the customer purchased, and Monetary measures the total value of the customer's purchases.

The latest transaction in the cleaned dataset occurred on 9 December 2011 at 12:50:00. A reference date one day after the latest transaction was used for Recency calculation.

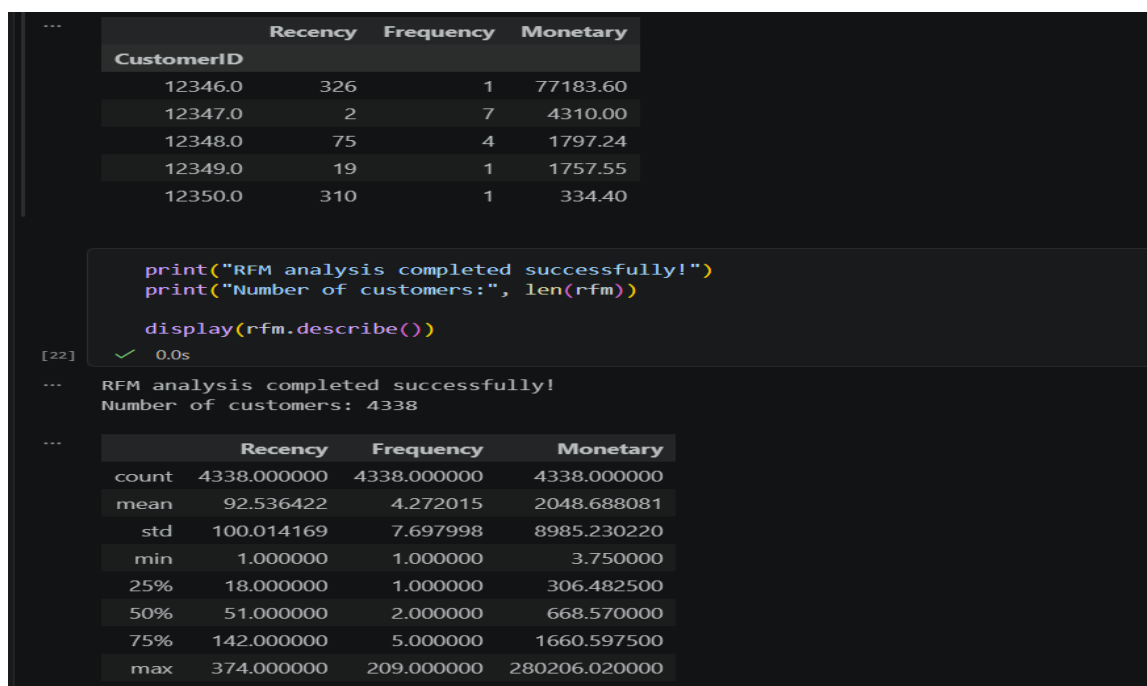


Figure 3.5: RFM Analysis – Recency, Frequency and Monetary Values

Interpretation:

RFM analysis was used to summarize customer purchasing behavior through **Recency**, **Frequency**, and **Monetary** measures. Recency indicates how recently a customer purchased, Frequency represents how often the customer purchased, and Monetary represents the total value of purchases. These three measures provide the behavioral features required for customer segmentation.

3.6 Feature Standardization

Recency, Frequency, and Monetary have different numerical scales. To prevent a feature with a larger numerical range from dominating the clustering process, the three RFM variables were standardized using StandardScaler.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_features = scaler.fit_transform(clustering_features)

scaled_features = pd.DataFrame(
    scaled_features,
    columns=clustering_features.columns,
    index=clustering_features.index
)

print("Features standardized successfully!")

display(scaled_features.head())
```

[24] ✓ 0.0s

... Features standardized successfully!

...

	Recency	Frequency	Monetary
CustomerID			
12346.0	2.334574	-0.425097	8.363010
12347.0	-0.905340	0.354417	0.251699
12348.0	-0.175360	-0.035340	-0.027988
12349.0	-0.735345	-0.425097	-0.032406
12350.0	2.174578	-0.425097	-0.190812

Figure 3.6: RFM Feature Standardization Using StandardScaler

Interpretation:

The Recency, Frequency, and Monetary variables have different numerical ranges, so they were standardized using StandardScaler. Standardization ensures that features with larger numerical values do not disproportionately influence the K-Means clustering process. The resulting standardized RFM features were then suitable for clustering analysis.

3.7 Elbow Method & K-Means Clustering

The Elbow Method was applied by fitting K-Means models for K values from 2 through 10 and recording the inertia for each value. The resulting curve was used to select a practical cluster count. Based on the elbow analysis, four clusters were selected for the final segmentation.

```

from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

K_range = range(2, 11)

for k in K_range:
    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )
    kmeans.fit(scaled_features)
    inertia.append(kmeans.inertia_)

plt.figure(figsize=(10, 6))

plt.plot(K_range, inertia, marker='o')

plt.title('Elbow Method for Optimal Number of Clusters')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('Inertia')
plt.xticks(K_range)
plt.grid(True)
plt.tight_layout()

plt.show()

```

✓ 2.4s

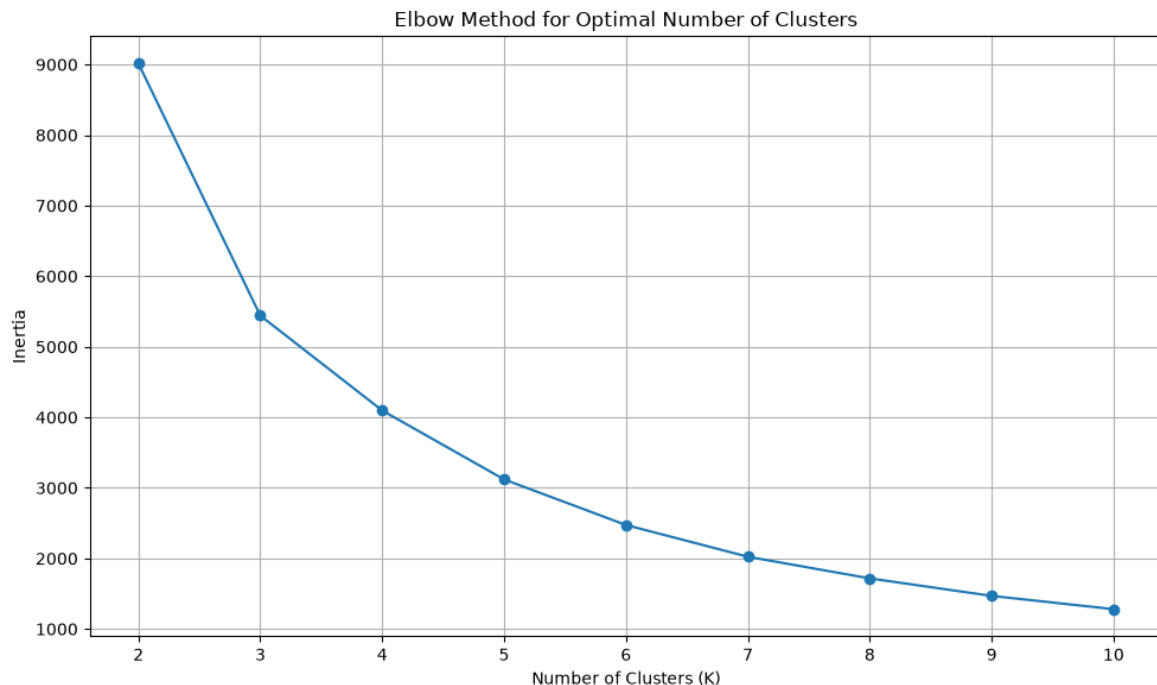


Figure 3.7: Elbow Method for Optimal Number of Clusters

Interpretation

The Elbow Method was evaluated for cluster values from **K = 2 to K = 10**. The inertia decreased substantially at the lower values of K and then showed a slower rate of improvement. Based on this pattern, **four clusters** were selected for the final K-Means model.

3.8 Cluster Visualization

Two scatter plots were generated to visualize the customer clusters. The first compares Recency with Monetary value, while the second compares Purchase Frequency with Monetary value. Cluster labels are represented through different plotted groups.

```
plt.figure(figsize=(10, 6))

plt.scatter(
    rfm['Recency'],
    rfm['Monetary'],
    c=rfm['Cluster'],
    alpha=0.6
)

plt.title('Customer Segmentation: Recency vs Monetary')
plt.xlabel('Recency (Days)')
plt.ylabel('Monetary Value')
plt.grid(True)
plt.tight_layout()

plt.show()
```

[28] ✓ 0.1s

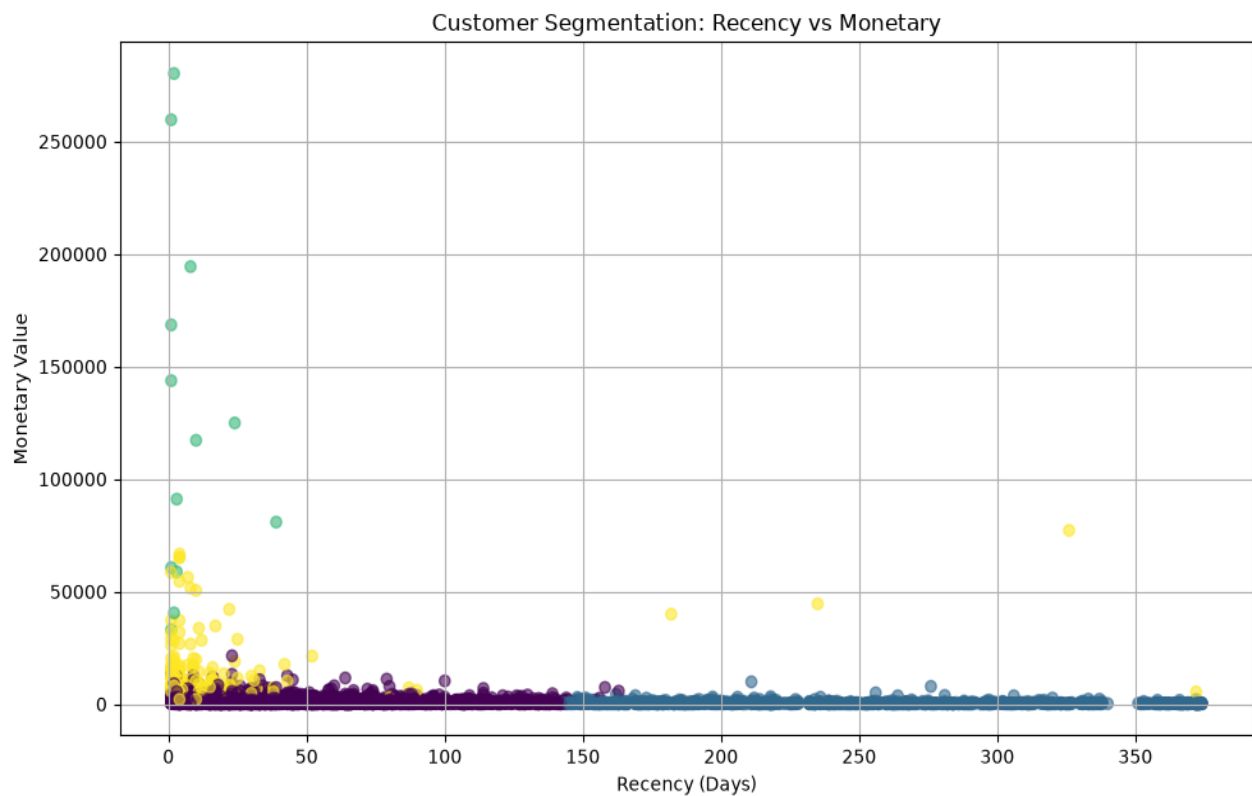


Figure 3.8: Customer Segmentation – Recency vs Monetary

Interpretation

The Recency versus Monetary visualization shows different customer purchasing patterns across the identified clusters. Customers with low recency and high monetary values represent recently active and valuable customers, while customers with higher recency and relatively lower monetary values indicate less recent and lower-value purchasing behavior. The visualization therefore supports the identification of distinct customer groups based on purchasing activity and value.

3.8 Cluster Visualization (continued)

```
plt.figure(figsize=(10, 6))

plt.scatter(
    rfm['Frequency'],
    rfm['Monetary'],
    c=rfm['Cluster'],
    alpha=0.6
)

plt.title('Customer Segmentation: Frequency vs Monetary')
plt.xlabel('Purchase Frequency')
plt.ylabel('Monetary Value')
plt.grid(True)
plt.tight_layout()

plt.show()
```

[29] ✓ 0.1s

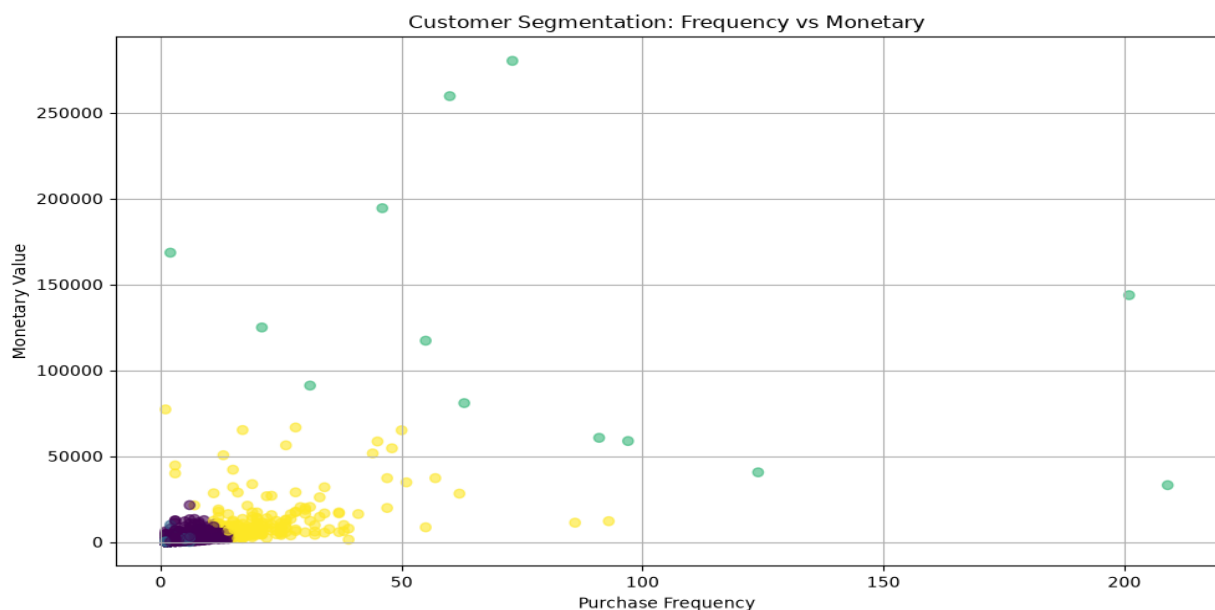


Figure 3.8: Customer Segmentation – Frequency vs Monetary

Interpretation

The Frequency versus Monetary visualization shows a small group of customers with exceptionally high purchase frequency and very high monetary value. Other customer groups are concentrated at lower frequency and spending levels. This separation supports the identification of different customer-value segments and helps distinguish high-value customers from regular purchasing groups.

3.9 Customer Cluster Profile

Cluster profiling was performed by calculating the mean Recency, Frequency, and Monetary values for each cluster. Customer counts were then added to the profile to compare the size and behavior of each group.

Cluster	Recency	Frequency	Monetary	Customer Count
0	43.70	3.68	1,353.63	3,054
1	248.08	1.55	478.85	1,067
2	7.38	82.54	127,187.96	13
3	15.50	22.33	12,690.50	204

Figure 3.9: Customer Cluster Profile

Interpretation

The cluster profile highlights clear behavioral differences among the four customer groups. Cluster 1 has the highest average recency of 248.08 days and the lowest average frequency and monetary value, indicating inactive behavior. In contrast, Cluster 2 has the lowest recency of 7.38 days, the highest frequency of 82.54, and the highest monetary value of 127,187.96, identifying it as the most valuable customer group.

3.10 Customer Segment Summary

The four clusters were assigned business-oriented names based on their average RFM characteristics. Cluster 0 was named Regular Customers, Cluster 1 At-Risk / Inactive Customers, Cluster 2 VIP / High-Value Customers, and Cluster 3 Loyal High-Value Customers.

```

segment_names = {
    0: 'Regular Customers',
    1: 'At-Risk / Inactive Customers',
    2: 'VIP / High-Value Customers',
    3: 'Loyal High-Value Customers'
}

rfm['Segment'] = rfm['Cluster'].map(segment_names)

segment_summary = rfm.groupby('Segment').agg(
    Customer_Count=('Segment', 'count'),
    Avg_Recency=('Recency', 'mean'),
    Avg_Frequency=('Frequency', 'mean'),
    Avg_Monetary=('Monetary', 'mean')
).round(2)

segment_summary

```

[32] ✓ 0.0s

Segment	Customer Count	Avg Recency	Avg Frequency	Avg Monetary
At-Risk / Inactive Customers	1,067	248.08	1.55	478.85
Loyal High-Value Customers	204	15.50	22.33	12,690.50
Regular Customers	3,054	43.70	3.68	1,353.63
VIP / High-Value Customers	13	7.38	82.54	127,187.96

The segment summary was successfully saved as customer_segment_summary.csv in the output folder.

Figure 3.10: Customer Segment Summary

Interpretation

The four clusters were converted into business-oriented customer segments based on their RFM characteristics. The Regular Customers segment contains 3,054 customers, while the At-Risk / Inactive Customers segment contains 1,067 customers. The Loyal High-Value Customers segment contains 204 customers, and the VIP / High-Value Customers segment contains only 13 customers but has the highest average frequency and monetary value. The summarized results were successfully exported to customer_segment_summary.csv.

3.11 Customer Count by Segment

The customer-count analysis was performed to compare the distribution of customers across the four identified segments. Regular Customers represent the largest segment with 3,054 customers, followed by At-Risk / Inactive Customers with 1,067 customers. Loyal High-Value Customers account for 204 customers, while VIP / High-Value Customers form the smallest segment with only 13 customers. The results indicate that most customers belong to the regular and at-risk groups, whereas the VIP segment is small but represents highly valuable customers.

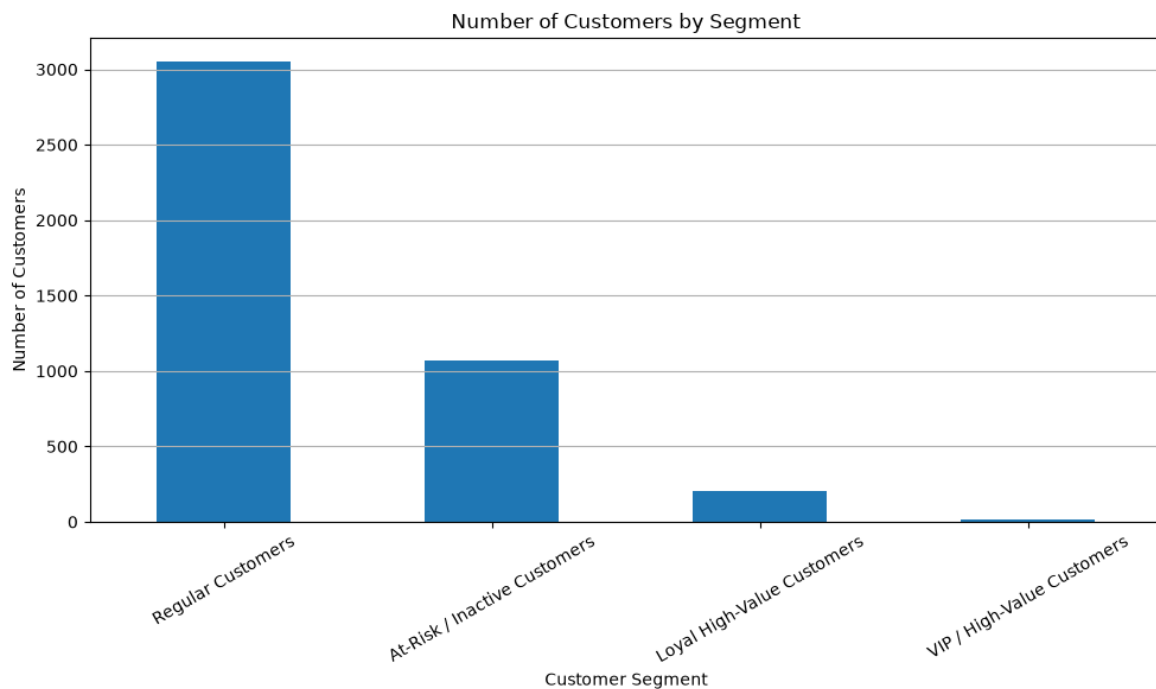


Figure 3.4: Number of Customers by Segment

Interpretation

The customer-count chart shows that Regular Customers form the largest segment with 3,054 customers, followed by At-Risk / Inactive Customers with 1,067 customers. Loyal High-Value Customers account for 204 customers, while VIP / High-Value Customers form the smallest group with only 13 customers. This indicates that the majority of customers belong to regular or at-risk groups, while the highest-value segment is comparatively small.

3.12 Marketing Recommendations by Customer Segment

The project includes segment-specific marketing recommendations based directly on the RFM characteristics of each group.

At-Risk / Inactive Customers have the highest average recency and the lowest purchase frequency and monetary value. Recommended actions include re-engagement campaigns, personalized discounts, reminder emails, and limited-time offers to encourage repeat purchases.

Regular Customers form the largest segment and show moderate purchase frequency, recent activity, and moderate spending. Recommended actions include loyalty programs, product recommendations, cross-selling, and personalized promotional campaigns.

Loyal High-Value Customers purchase frequently, have recent activity, and generate substantially higher monetary value than regular customers. Recommended actions include VIP benefits, early product access, personalized recommendations, loyalty rewards, and exclusive offers.

VIP / High-Value Customers are a small but extremely valuable group with very recent purchases, exceptionally high purchase frequency, and the highest monetary value. Recommended actions include premium customer service, exclusive offers, personalized communication, priority support, and retention-focused relationship management.

```
## Marketing Recommendations by Customer Segment

### 1. At-Risk / Inactive Customers
These customers have the highest average recency and the lowest purchase frequency and monetary value. They have not purchased recently and may be at risk of becoming inactive.

**Recommended Action:** Use re-engagement campaigns, personalized discounts, reminder emails, and limited-time offers to encourage repeat purchases.

### 2. Regular Customers
This is the largest customer segment. These customers show moderate purchase frequency, recent activity, and moderate spending.

**Recommended Action:** Encourage repeat purchases through loyalty programs, product recommendations, cross-selling, and personalized promotional campaigns.

### 3. Loyal High-Value Customers
These customers purchase frequently, have recent activity, and generate substantially higher monetary value than regular customers.

**Recommended Action:** Strengthen customer loyalty through VIP benefits, early access to products, personalized recommendations, loyalty rewards, and exclusive offers.

### 4. VIP / High-Value Customers
This is a small but extremely valuable customer segment. These customers have very recent purchases, exceptionally high purchase frequency, and the highest monetary value.

**Recommended Action:** Provide premium customer service, exclusive offers, personalized communication, priority support, and retention-focused relationship management to protect this high-value segment.

### Overall Insight
The customer segmentation analysis identifies four distinct behavioral groups. The business should focus on reactivating at-risk customers, developing regular customers into loyal customers, retaining loyal high-value customers, and providing personalized treatment to VIP customers.
```

Figure 3.5: Marketing Recommendations by Customer Segment

Interpretation

The segmentation results support different marketing strategies for each customer group. At-Risk / Inactive Customers require re-engagement campaigns, while Regular Customers can be encouraged through loyalty programs and cross-selling. Loyal High-Value Customers should receive loyalty rewards and exclusive benefits, whereas VIP / High-Value Customers require personalized communication, premium service, and retention-focused strategies.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Results Obtained

The Customer Segmentation Analysis was successfully implemented in Python using Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The dataset was cleaned, customer-level metrics were generated, RFM features were calculated and standardized, the Elbow Method was applied, and K-Means clustering produced four customer segments.

The analysis produced five documented visual outputs: the Elbow Method, Recency versus Monetary scatter plot, Frequency versus Monetary scatter plot, customer count by segment chart, and marketing recommendations evidence. A customer segment summary was also exported as `customer_segment_summary.csv`.

4.2 Key Findings

1. The original Online Retail dataset contains 541,909 records and 8 variables.
2. 5,268 duplicate records were identified and removed.
3. 9,251 cancelled transactions were identified and removed.
4. 1,336 records had invalid non-positive Quantity values and 2,512 had invalid non-positive UnitPrice values.
5. After removing transactions without CustomerID, 392,692 customer-linked transaction records remained.
6. Customer-level analysis produced 4,338 unique customers.
7. Four customer clusters were selected using the Elbow Method and implemented using K-Means.
8. Regular Customers are the largest segment with 3,054 customers.
9. At-Risk / Inactive Customers contain 1,067 customers and have the highest average recency.
10. Loyal High-Value Customers contain 204 customers and show high purchase frequency and monetary value.
11. VIP / High-Value Customers contain only 13 customers but have the highest average frequency of 82.54 and average monetary value of 127,187.96.

4.2 Business Recommendations

1. Use re-engagement campaigns, personalized discounts, reminder emails, and limited-time offers for At-Risk / Inactive Customers.
2. Use loyalty programs, product recommendations, cross-selling, and personalized promotions to increase the value of Regular Customers.
3. Strengthen retention of Loyal High-Value Customers using VIP benefits, early access, personalized recommendations, loyalty rewards, and exclusive offers.
4. Provide premium service, priority support, personalized communication, and retention-focused relationship management for VIP / High-Value Customers.
5. Use the segment summary as a basis for targeted marketing rather than applying the same campaign to every customer.

4.3 Conclusion

The Level 1 – Task 2 Customer Segmentation Analysis project successfully transformed raw online retail transactions into meaningful customer behavioral segments. The implementation demonstrated a complete workflow from data loading and cleaning through customer aggregation, RFM analysis, feature standardization, K-Means clustering, visualization, profiling, and business recommendations.

The results show that customer behavior differs substantially across the four identified groups. Regular Customers form the majority of the customer base, At-Risk / Inactive Customers require re-engagement, Loyal High-Value Customers require retention and loyalty strategies, and VIP / High-Value Customers represent a very small but exceptionally valuable group.

Overall, the project demonstrates how customer segmentation can support targeted marketing, customer retention, loyalty development, and more personalized business decision-making.

4.4 Future Enhancements

1. Evaluate clustering quality using additional metrics such as silhouette score.
2. Experiment with alternative clustering methods such as hierarchical clustering or DBSCAN.
3. Develop an interactive Power BI or Streamlit dashboard for customer segment exploration.
4. Track segment migration over time to identify customers moving from regular to loyal or from loyal to at-risk behavior.
5. Integrate product-level preferences into the segments for more detailed recommendation strategies.
6. Automate the segmentation pipeline so that new transactions can update customer segments regularly.

7. Develop predictive models for customer churn, repeat purchase probability, and customer lifetime value.

4.5 References

1. Project-provided Online Retail.xlsx dataset used in the Level 1 – Task 2 notebook.
2. Python Pandas documentation and library usage as implemented in the project notebook.
3. Scikit-learn StandardScaler and K-Means functionality as implemented in the project notebook.
4. Matplotlib and Seaborn visualization libraries as implemented in the project notebook.
5. Level 1 – Task 2: Customer Segmentation Analysis project requirements.